

AI Resume Screening & Interview Assistant - Project Documentation

Project Type: Full Stack Generative AI Application

Difficulty: Intermediate to Advanced

1. Project Overview

Build an AI-powered recruitment platform that automates resume screening, ATS scoring, candidate ranking, interview question generation, AI interviews, transcript generation, and candidate evaluation.

2. Business Problem

Recruiters spend significant time reviewing resumes manually. This system reduces screening effort and improves hiring efficiency using AI.

3. Core Features

- User Authentication (JWT)
- Resume Upload
- Job Description Upload
- ATS Score Generation
- Skill Gap Analysis
- Candidate Ranking
- AI Interview Question Generator
- Voice Interview Assistant
- Interview Transcript Generation
- Candidate Evaluation Dashboard

4. Recommended Tech Stack

Frontend: React, Tailwind CSS

Backend: FastAPI

Database: PostgreSQL

AI: OpenAI/Gemini, LangChain

Vector DB: ChromaDB/FAISS

Speech: Whisper, TTS

Deployment: Docker, AWS, Render, Vercel

5. System Architecture

Frontend -> FastAPI Backend -> PostgreSQL

Resume/JD Processing -> Embeddings -> Vector DB

LLM -> ATS Scoring, Ranking, Interview Questions, Evaluation

6. Database Design

Users, Jobs, Candidates, Resumes, Interviews, Transcripts, Evaluations tables.

7. AI Modules

Resume Parser, ATS Scorer, Skill Gap Analyzer, Candidate Ranker, Interview Generator, Voice Interview Agent.

8. APIs

/register

/login

/upload-resume

/upload-job-description

/ats-score

/rank-candidates

/generate-interview-questions

/start-interview

/transcript

/evaluation

9. Advanced Enhancements

Multi-Agent Hiring Workflow, Email Notifications, Calendar Integration, HR Dashboard, Analytics.

10. Interview Discussion Points

Explain architecture, RAG usage, embeddings, candidate ranking logic, prompt engineering, deployment strategy, scalability, and security.